Lecture 1

Course Introduction & Supervised Learning Basics

ECON6083: Machine Learning for Economists

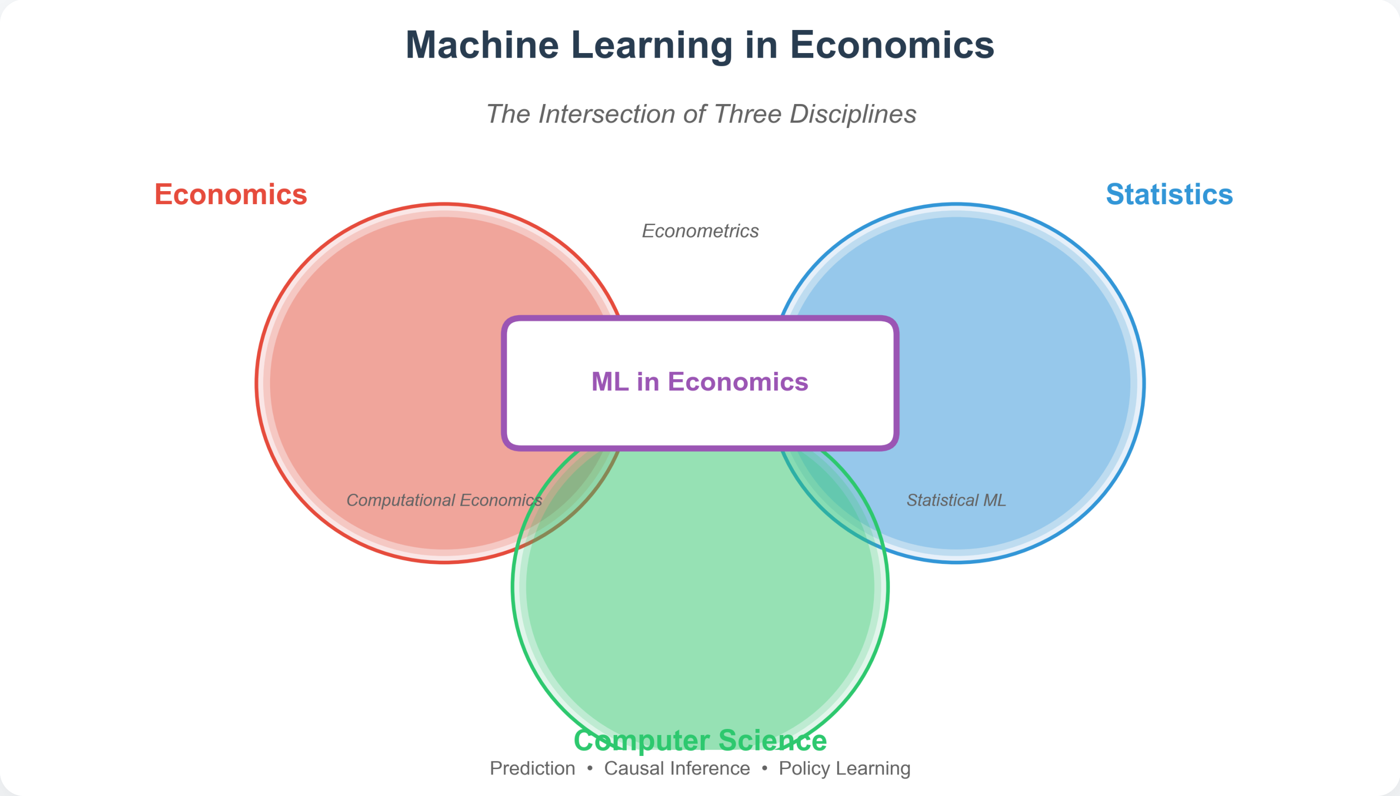
Today's Roadmap

#	TOPIC	FOCUS
1	The Historic Convergence	Econometrics meets ML; Why now?
2	Three Pillars	Prediction, Causal Inference, Policy Learning
3	Course Structure	Assessment, projects
4	From OLS to Prediction	Paradigm shift; Bias-variance tradeoff
5	Bias-Variance Decomposition	Components and tradeoff

Part 1

The Historic Convergence of Two Fields

Two Parallel Rivers



Econometrics (Economics + Statistics)

- Focus: **Causal Inference** — estimating $\hat{\beta}$
- Questions: "What is the return to education?"
- Priority: Unbiasedness, interpretability

Machine Learning (Computer Science)

- Focus: **Prediction** — forecasting $\hat{y}$
- Questions: "Will this user click the ad?"

The Big Data Era: Rivers Converge

Why the convergence?

Economists realized:

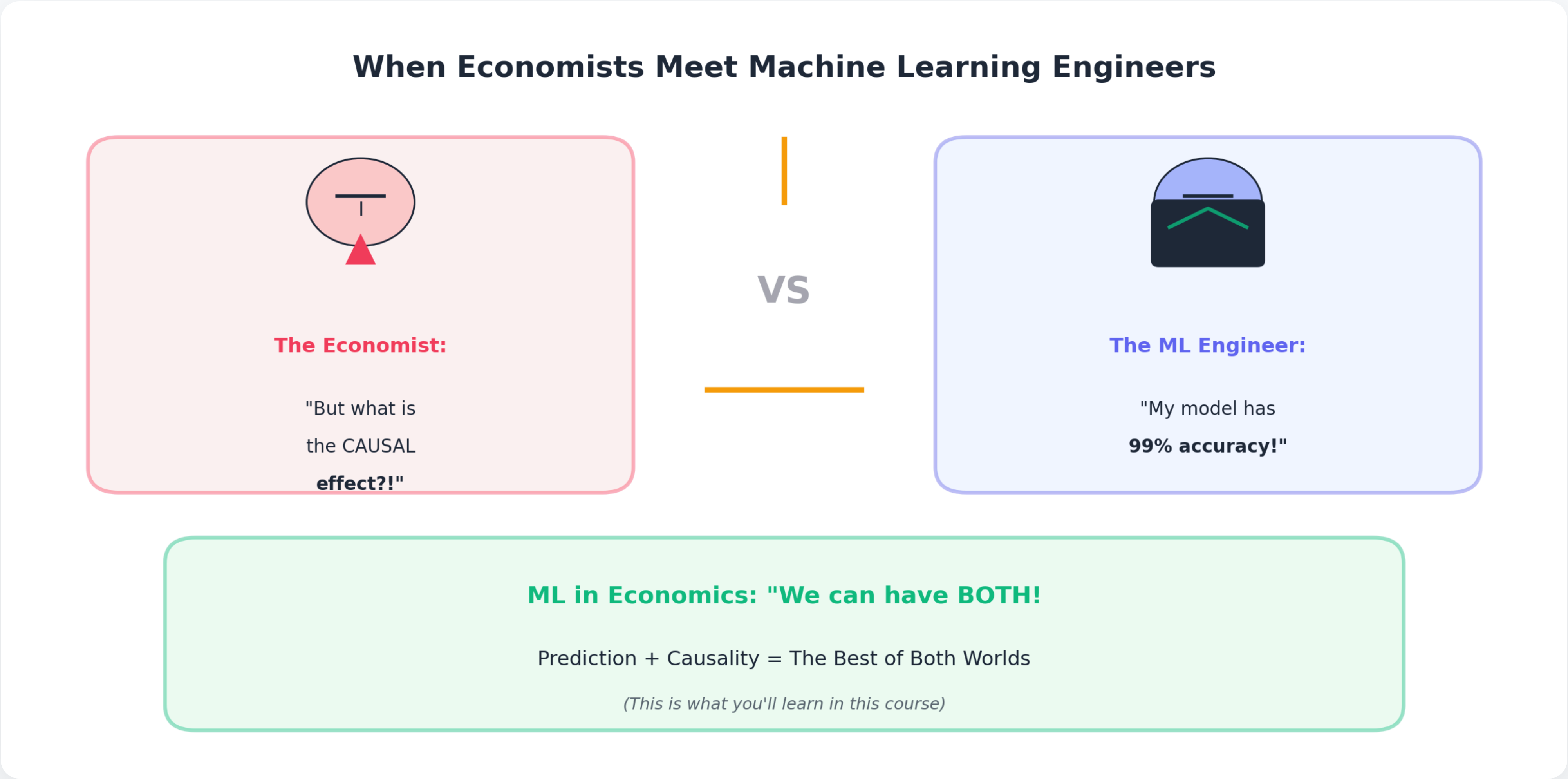
- Many decisions rely on high-quality predictions
- Big data offers unprecedented opportunities

Computer scientists realized:

- Pure prediction fails under policy interventions
- Causal reasoning is essential (Lucas Critique revisited)

Result: A new interdisciplinary field emerges

Economists vs Machine Learning Engineers



Part 2

The Map of ML Applications

Three Pillars of ML in Economics

1. Pure Prediction

- Use ML for forecasting
- Example: Poverty mapping with satellites

2. Causal Inference

- ML as a tool for better causal estimates
- Example: Double ML for high-dimensional controls

3. Policy Learning

- Optimal heterogeneous treatment assignment
- Example: Personalized donation suggestions

Pillar 1

Pure Prediction

Pure Prediction: Core Logic

Key insight: We care about whether X predicts Y, not whether X causes Y

If a variable improves out-of-sample prediction, it has value

- No need to establish causality
- Focus on R^2 and forecast accuracy
- Embrace "black box" models if they work

Case Study: Satellite Imagery & Poverty

The Problem: Reliable GDP data is scarce in developing countries

- Household surveys are expensive
- Data is infrequent and unreliable

The ML Solution: Use satellite imagery

- Nighttime lights, roofing materials, roads
- Convolutional Neural Networks (CNNs)
- Consumption r^2 up to 0.75 (Jean et al., 2016)

Case Study: Predicting Flight Risk

The Problem: Judges must decide pretrial detention

- Who will flee if released?
- Human judgment is inconsistent

The ML Solution (Kleinberg et al. 2018):

- Use criminal records + demographics
- Train algorithms to predict flight risk
- Algorithms outperform human judges

Impact: Could reduce pretrial detention by 40% without increasing crime (Kleinberg, Lakkaraju, Leskovec, Ludwig & Mullainathan, 2018, *QJE*).

Pillar 2

Causal Inference with ML

The High-Dimensional Control Problem

Traditional challenge: Controlling for confounders

Example: Estimating returns to education

- Need to control for "ability"
- Ability has thousands of dimensions
- Resume text, social networks, test scores, etc.

When $p \approx n$: Traditional OLS fails

Solution: Double/Debiased Machine Learning

Proposed by Chernozhukov et al.

Key idea:

- Use flexible ML models (Random Forests, LASSO)
- Fit non-linear relationships between controls and outcomes
- "Partial out" confounding factors
- Estimate causal parameter on residuals

Result: ML flexibility + Econometric unbiasedness

Pillar 3

Policy Learning

From Average to Personalized Effects

Traditional question: "Is this treatment effective on average?"

- Average Treatment Effect (ATE)
- One-size-fits-all policy

Policy Learning question: "Who should receive this treatment?"

- Heterogeneous Treatment Effects (HTE)
- Personalized intervention
- Maximize social welfare

Case Study: Personalized Donations

PayPal-Athey collaboration study

Discovery: Users respond very differently to donation suggestions

- Older/high-income users: tolerate high suggestions
- New users: deterred by high asks
- Middle group: responsive to moderate amounts

Traditional approach: One fixed default for everyone (suboptimal)

Personalized Donations: The ML Solution

Use Causal Forests to learn optimal strategy:

- Estimate individual-level treatment effects
- Assign different default amounts by user type
- Maximize total donations

Results:

- Pareto improvement over any fixed strategy
- Win-win: more money raised, better user experience

Part 3

Course Structure

Assessment Structure

COMPONENT	WEIGHT	DESCRIPTION
In-Class Exercises (x10)	10%	1% each, completion-based
Assignments (x4)	50%	4 assignments, 12.5% each
Final Project	40%	Empirical research report

Total: 100 points

In-Class Exercises (10%)

- **One exercise per lecture** (10 total, 1% each)
- Submit your completed notebook **by end of day**
- Graded on **completion**, not correctness
- Must run without errors to receive full credit
- Late submissions: 0 marks (no exceptions)

Assignments (50%)

Theoretical Derivation (40%):

- Manually derive closed-form solutions
- Example: Ridge Regression, Bias-Variance decomposition
- Understand the "why" behind algorithms

Programming (60%):

- Implement "From Scratch"
- Example: Hand-code K-Fold Cross-Validation
- Understand data leakage risks

Academic Integrity

All assignments will be checked for plagiarism

- **Code**: Write independently — no copy-paste from classmates, online, or AI tools
- **Report**: All writing must be your own words with proper citations
- **Consequences**: Plagiarism = **zero marks** and Faculty reporting

Final Project (40%)

Two tracks (choose one):

Track 1: Replication & Extension

- Choose ML paper from top journal (AER, QJE, JEP)
- Published within 3 years; replicate main results
- Extend with new algorithms, data, or robustness tests

Track 2: Critical Review

- Read 15-20 papers on a specific application (e.g., "ML in Antitrust")
- Identify field evolution and limitations; propose future directions

Deliverable: Research report (15-20 pages) — literature review, methodology, empirical results, discussion

Part 4

From OLS to Prediction

Traditional Econometrics: The OLS Framework

The classic linear model:

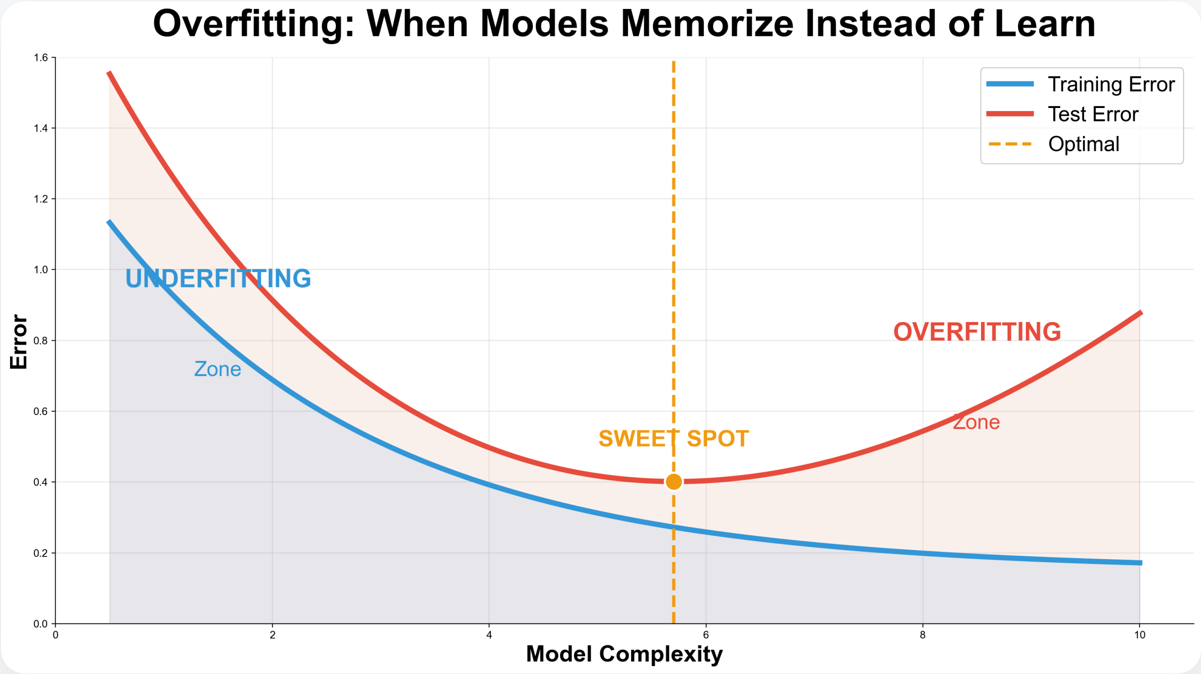
$$Y = X\beta + \varepsilon$$

Under classical assumptions:

- OLS is BLUE (Best Linear Unbiased Estimator)
- $E[\hat{\beta}] = \beta$ (unbiasedness)
- Minimum variance among unbiased estimators

Unbiasedness is non-negotiable in traditional framework

The Problem: OLS Excels in Training, Fails in Testing



Traditional focus: Minimize in-sample error — adding variables always decreases R^2_{train}, a 99-degree polynomial achieves perfect fit.

Reality: Model memorizes noise, terrible out-of-sample performance.

ML's Paradigm Shift

Accept small bias for lower variance

Key insight:

- Perfect unbiasedness is not the goal
- What matters: Out-of-sample prediction accuracy
- Biased estimators can predict better

New focus:

- Training error: How well we fit the data
- Test error: How well we predict new data
- **Test error is what matters!**

Two Types of Error

Training Error:

$$MSE_{train} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

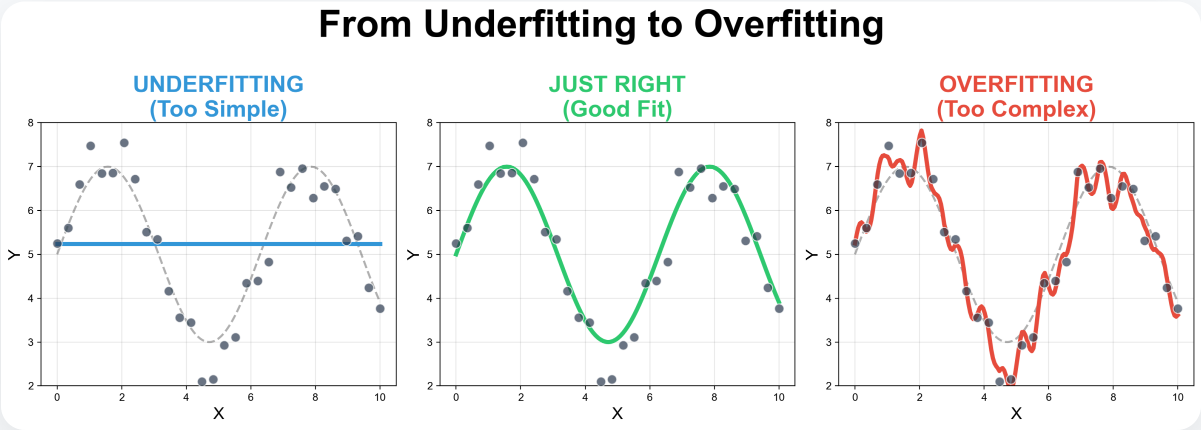
- Measured on data used to fit the model
- Can always be reduced by adding complexity

Test Error:

$$MSE_{test} = E[(y_{new} - \hat{f}(x_{new}))^2]$$

- Measured on new, unseen data
- The true measure of model quality

Overfitting: Memorizing vs. Learning

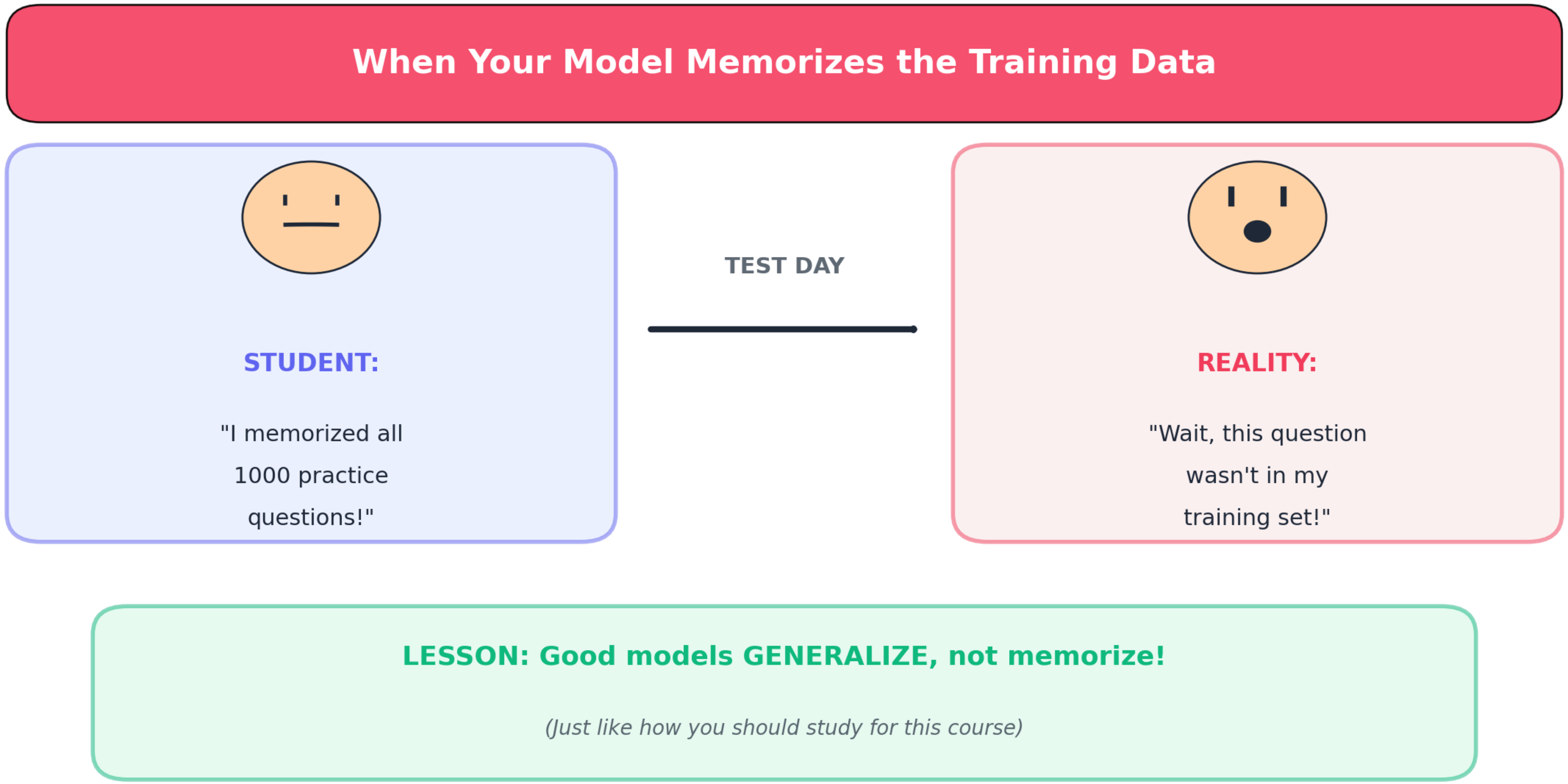


Signal: True relationship between X and Y (e.g., Education → wages)

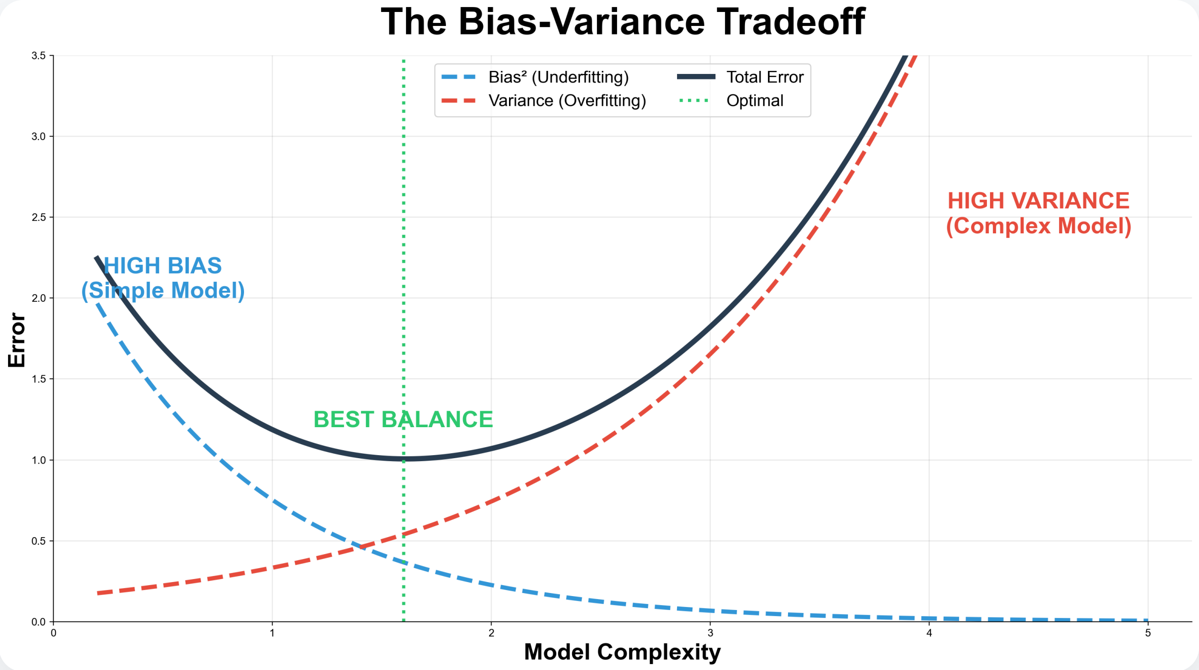
Noise: Random fluctuations (luck, measurement error)

Overfitted model: Memorizes both — fits training data perfectly but makes wildly wrong predictions on new data

A Familiar Story...



The U-Shaped Curve



Model Complexity (x-axis) vs. **Error** (y-axis): Training Error decreases monotonically. Test Error is U-shaped — first captures signal, then memorizes noise. **Sweet spot**: Minimum of test error curve.

Data Splitting Protocol

Critical Rule: Separate train and test

Training Set (70-80%):

- Used to estimate model parameters
- All model fitting happens here

Test Set (20-30%):

- Locked away until final evaluation
- Simulate "new" unseen data
- **Never** use for model selection!

Data Leakage: The Cardinal Sin

What is data leakage?

- Using test set information during training
- Adjusting model after seeing test results
- Standardizing before splitting data

Why it's fatal:

- Test error no longer measures true performance
- Model fails in real deployment

Golden rule: Test set is sacred

Part 5

Bias-Variance Decomposition

The Mathematical Foundation

Expected Prediction Error at point x_0:

$$\text{EPE}(x_0) = \underbrace{(E[\hat{f}(x_0)] - f(x_0))^2}_{\text{Bias}^2} + \underbrace{E[(\hat{f}(x_0) - E[\hat{f}(x_0)])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}}$$

Three components determine prediction quality

Component 1: Bias

Definition: Systematic error from model oversimplification

Causes:

- Model too simple for true relationship
- Example: Linear model for non-linear data

Consequence: Persistent prediction errors in same direction

Component 2: Variance

Definition: Sensitivity to training data fluctuations

Causes:

- Model too complex, memorizes noise
- Slight data changes → wild prediction swings

Consequence: Unstable predictions, poor generalization

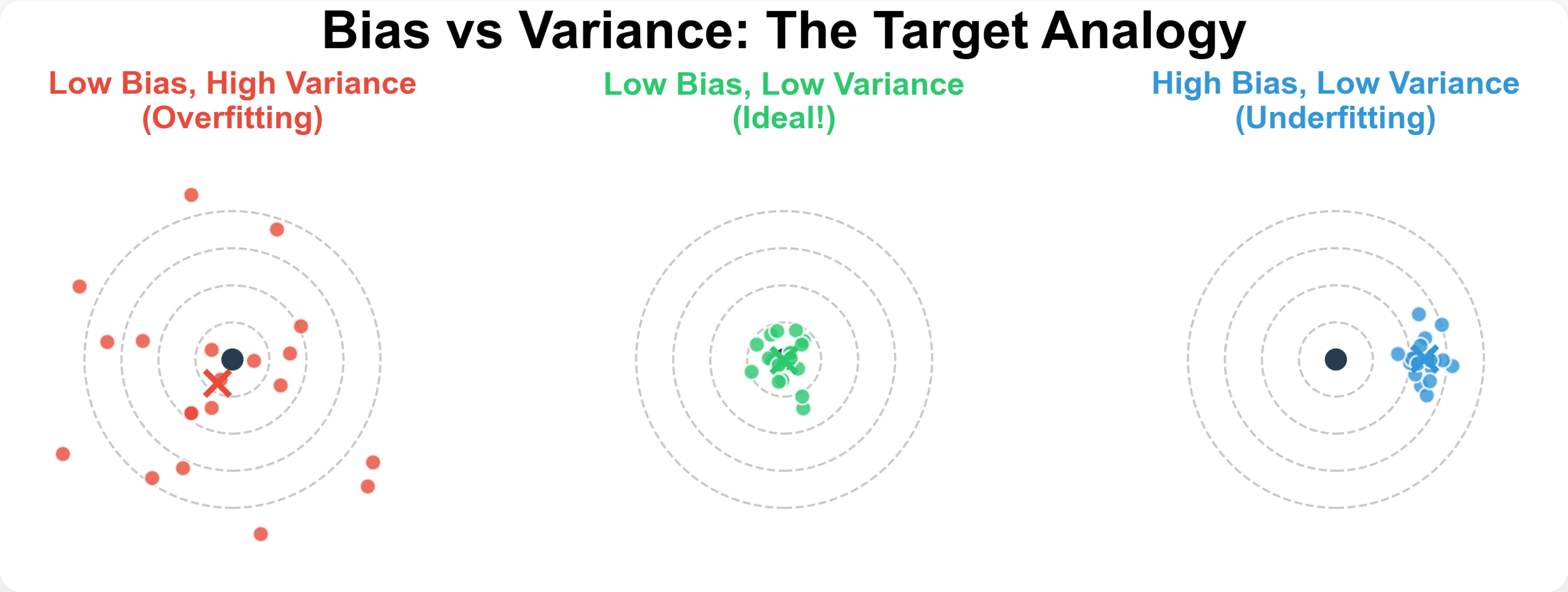
Component 3: Irreducible Error

Definition: Inherent noise in the data (σ^2)

- Cannot be reduced through better modeling
- Represents true randomness in the system
- Theoretical upper limit of accuracy

Example: Random measurement error, true stochasticity

The Bias-Variance Tradeoff



Low Bias, Low Variance: Tightly clustered on bullseye (ideal)

High Bias, Low Variance: Tightly clustered but off-target

Low Bias, High Variance: Scattered around bullseye

High Bias, High Variance: Scattered and off-target (disaster)

Model Complexity and the Tradeoff

Increasing Complexity:

- Bias ↓ (captures more patterns)
- Variance ↑ (memorizes more noise)

Decreasing Complexity:

- Bias ↑ (misses patterns)
- Variance ↓ (stable predictions)

Goal: Find the sweet spot minimizing total MSE

Why Biased Estimators Can Be Better

Traditional econometrics: Unbiasedness is sacred — $E[\hat{\beta}] = \beta$ always

ML insight: Total error matters, not just bias

- Small bias + much lower variance = better predictions
- Example: Ridge Regression

Philosophy shift:

- From "perfectly right on average"
- To "approximately right and stable"

Part 6

Summary and Looking Ahead

Key Paradigm Shifts

From Unbiasedness to Prediction:

- Accept small bias for lower variance
- Focus on out-of-sample performance

From Low-Dimensional to High-Dimensional:

- Embrace regularization; work with $p > n$

From Average Effects to Heterogeneity:

- Personalized policy learning
- Optimal treatment assignment

Key Takeaways

CONCEPT	KEY POINT
1. The Convergence	Economics + ML = powerful new toolkit
2. Three Pillars	Prediction, Causal Inference, Policy Learning
3. The Tradeoff	Bias vs. Variance
4. The Solution	Regularization for high dimensions
5. The Goal	Better decisions through better predictions

Common Pitfalls

1. Ignoring the bias-variance tradeoff

- Overfitting to training data; not using validation sets

2. Applying ML blindly to causal questions

- Prediction ≠ Causal inference; need explicit identification strategy

3. Forgetting economic interpretability

- Black-box models without explanation; SHAP and feature importance help

Further Reading

PAPER	AUTHORS	KEY CONTRIBUTION
Machine Learning: An Applied Econometric Approach	Mullainathan & Spiess (2017), *JEP*	ML for economists
The Impact of Machine Learning on Economics	Athey (2018), *NBER*	Comprehensive survey
Big Data: New Tricks for Econometrics	Varian (2014), *JEP*	Introduction to ML methods
Prediction Policy Problems	Kleinberg et al. (2015), *Science*	When prediction matters

Data and Code

This lecture:

- Slides source: `_slides/Lecture01-Introduction and Supervised Learning-Slides.md`
- Exercises: `exercises/Lecture01-Exercise.ipynb`

Key packages:

- Python: `scikit-learn`, `pandas`, `matplotlib`

Datasets:

- Pennsylvania reemployment bonus experiment (public)

Thank You!

Next Lecture: Regularization and High-Dimensional Regression

Reading: Mullainathan & Spiess (2017), Sections 1-3